

Results

Descriptives - AI Specialists

INDICATORS	N	Mean	SD	Verbal Interpretation
The AI model achieves sufficient detection accuracy for the three lung cancer classes in preliminary screening.	16	4.63	0.500	Excellent
The model produces consistent classification results when analyzing the same CT scan multiple times.	16	4.50	0.632	Excellent
Model Performance (Composite Mean)	16	4.56	0.479	Excellent
The AI integration into the web platform functions well for real-time CT scan analysis.	16	4.81	0.403	Excellent
The system processes medical imaging data efficiently.	16	4.56	0.814	Excellent
System Architecture (Composite Mean)	16	4.69	0.574	Excellent
The cross-dataset validation approach (training on Mohamed Kany, testing on TCIA) is methodologically sound.	16	4.63	0.619	Excellent
The 80% accuracy target aligns with preliminary screening objectives.	16	4.69	0.479	Excellent
Validation Methodology (Composite Mean)	16	4.66	0.507	Excellent
The security measures protect sensitive medical imaging data effectively.	16	4.50	0.730	Excellent
The platform architecture can scale for clinical deployment.	16	4.50	0.730	Excellent
Scalability and Security (Composite Mean)	16	4.50	0.606	Excellent

Legend:

- 1.00 - 1.49 = Very Poor
- 1.50 – 2.49 = Poor
- 2.50 – 3.49 = Good
- 3.50 – 4.49 = Very Good
- 4.50 – 5.00 = Excellent

Descriptives - Medical Experts

INDICATORS	N	Mean	SD	Verbal Interpretation
The AI system identifies lung abnormalities effectively for lung cancer detection	16	4.00	0.816	Very Good
The three-class categorization (adenocarcinoma, squamous cell carcinoma, normal) provides clinically meaningful information for case prioritization.	16	4.06	0.772	Very Good
Clinical Accuracy (Composite Mean)	16	4.03	0.694	Very Good
The platform design improves coordination in lung cancer detection workflows	16	4.25	0.856	Very Good
The communication features facilitate effective patient-physician interaction during consultation.	16	4.00	0.816	Very Good
Clinical Workflow Integration (Composite Mean)	16	4.13	0.742	Very Good
The platform clearly positions AI findings as preliminary analysis that supplements physician expertise without replacing clinical judgment.	16	4.13	0.806	Very Good
The medical terms and explanations provided for preliminary findings are understandable and appropriate	16	4.44	0.727	Very Good
Clinical Decision Support (Composite Mean)	16	4.28	0.682	Very Good
This platform reduces the 2–4-week delay between CT acquisition and physician review.	16	3.94	0.854	Very Good
This proof-of-concept platform supports coordination and case prioritization in lung cancer detection workflows.	16	4.25	0.775	Very Good
Clinical Utility (Composite Mean)	16	4.25	0.775	Very Good

Legend:

1.00 - 1.49 = Very Poor

1.50 – 2.49 = Poor

2.50 – 3.49 = Good

3.50 – 4.49 = Very Good

4.50 – 5.00 = Excellent

Descriptives - Others

INDICATORS	N	Mean	SD	Verbal Interpretation
The web application is easy to navigate.	22	4.77	0.429	Excellent
Uploading CT scans and accessing results is straightforward.	22	4.82	0.501	Excellent
Ease of Use	22	4.80	0.367	Excellent
The findings and medical terms are explained clearly	22	4.32	0.839	Very Good
The visual indicators (labels, confidence scores) are easy to understand.	22	4.55	0.671	Excellent
Understanding Results	22	4.43	0.729	Very Good
The platform improves communication with my healthcare provider.	22	4.64	0.658	Excellent
The result presentation helps me understand my case status while waiting for physician review.	22	4.55	0.671	Excellent
Communication	22	4.59	0.610	Excellent
The completeness of information provided in the analysis results is satisfactory.	22	4.50	0.673	Excellent
The platform provides the information I need to understand my lung cancer detection results.	22	4.82	0.395	Excellent
Satisfaction	22	4.66	0.447	Excellent

Legend:

1.00 - 1.49 = Very Poor

1.50 - 2.49 = Poor

2.50 - 3.49 = Good

3.50 - 4.49 = Very Good

4.50 - 5.00 = Excellent

References

[1] The jamovi project (2024). *jamovi*. (Version 2.6) [Computer Software]. Retrieved from

<https://www.jamovi.org>.

[2] R Core Team (2024). *R: A Language and environment for statistical computing*. (Version 4.4)

[Computer software]. Retrieved from <https://cran.r-project.org>. (R packages retrieved from CRAN snapshot 2024-08-07).